

A Hybrid System Framework for Unified Impedance and Admittance Control

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Abstract Impedance Control and Admittance Control are two distinct implementations of the same control goal but their stability and performance characteristics are complementary. Impedance Control is better suited for dynamic interaction with stiff environments and Admittance Control is better suited for interaction with soft environments or operation in free space. In this paper, we use a hybrid systems framework to develop an entire family of controllers that have Impedance Control and Admittance Control at two ends of its spectrum; and intermediate controllers that have stability and performance characteristics that are an interpolation of those of Impedance Control and Admittance Control. The hybrid systems framework provides the scope for maintaining stability and achieving the best performance by choosing a specific

controller for a given environment and by continuously changing the controller to adapt to a changing environment. The advantage of our approach is demonstrated with an extensive case study of a one-dimensional system and through experiments with the joint of a lightweight robotic arm.

Keywords Impedance control · Admittance control · Hybrid control

1 Introduction

The most successful applications of industrial robots are primarily restricted to tasks where there is no dynamic interaction between the robot and its environment. These tasks, such as pick and place operations, spray-painting and welding, require accurate position control but there is virtually no energy exchange between the robot and its environment. Robotic manipulation tasks in which energy is exchanged with the environment through dynamic interaction has been a subject of considerable research and two fundamental control methodologies have been proposed. The first approach, known as “Hybrid Position and Force Control,” was developed by Raibert and Craig [25]. The compliance control methodology of Mason [20] is a variation of this approach. In hybrid position and force control, the task space is divided into position-controlled and force-controlled sub-spaces since both position and force cannot be controlled along any

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given direction. The position of the manipulator is controlled in the position-controlled subspace and contact forces with the environment are controlled in the force-controlled subspace. The hybrid position and force control methodology ignores the dynamic coupling between the manipulator and the environment. To address this problem, Hogan proposed “Impedance Control” [11–13], wherein the mechanical impedance of the manipulator is regulated to that of a target model. Impedance control can be seen as a generalization of stiffness control [27], where the control objective focuses on the static behavior. It establishes a dynamical relationship between the end-effector position and force and provides a unified framework for manipulator control in free-space and in compliant motion with environmental contact.

Efforts have been made to combine impedance control and hybrid position and force control. Anderson and Spong [3] proposed an inner/outer loop control strategy; the inner loop is based on feedback linearization with force cancelation and the outer loop is similar to the classical hybrid position and force controller [25] with impedance control in the position-controlled subspace. Liu and Goldenberg [18] proposed a robust hybrid controller with impedance control in the position-controlled subspace and desired inertia and damping in the force-controlled subspace to improve dynamic behavior. A PI controller was used to achieve robustness and the controller was implemented in a two-dof direct-drive robot.

There are two ways of implementing impedance control, depending on the causality of the controller. These are often referred to as “Impedance Control” and “Admittance Control” in the literature. Although both implementations were referred to as impedance control by Hogan [11], we make a distinction since it is central to the work presented here. In Impedance Control the controller is an impedance¹ and the manipulator is an admittance² and in Admittance Control the controller is an admittance and the manipulator is an impedance. The stability properties of Impedance and Admittance Control in the presence of non-ideal effects such as time delay was discussed by Lawrence [16]. It was shown that desired values of stiffness and damping become more restricted

as time delay increases. More importantly, systems with Impedance and Admittance control have opposite stability requirements: low stiffness can be achieved with low feedback gains with Impedance Control but requires high gains for Admittance Control. Similarly, high stiffness requires high gains for Impedance Control but can be implemented with low gains with Admittance Control.

In general, robotic systems with Impedance Control have stable dynamic interaction with stiff environments but have poor accuracy in free-space due to friction and other unmodeled dynamics. This problem can be mitigated using inner loop torque sensing/control or through hardware modifications such as low-friction joints and direct-drive actuators. Impedance Control has been implemented with inner loop torque sensing and control in DLR’s light-weight robot [21] and ATR’s³ Humanoid built by Sarcos⁴ [6]; and with low-friction joints and low inertia in the Phantom⁵ haptic device. In contrast to Impedance Control, Admittance Control provides high level of accuracy in non-contact tasks but can result in instability during dynamic interaction with stiff environments. This problem can be eliminated using series elastic actuation or compliant end-effectors but this reduces performance. In contrast to direct-drives which are used in conjunction with Impedance Control, Admittance Control requires high transmission ratios such as harmonic drives for precise motion control, and industrial robotic systems are good examples.

Several control approaches have been proposed for compensating the limitations of Impedance Control and Admittance Control. The effect of uncertainties in the robot model has been addressed in [19] based on the adaptation algorithm from [30]. For improving the robustness of Admittance Control, Aghili [1] proposed an approach based on robust inverse dynamics. In [15] Internal Model Control was applied for improving the position accuracy of an impedance controller.

Robotic systems with Impedance Control and Admittance Control have complementary advantages and disadvantages and neither one of them can provide optimal performance for the complete range of

¹A physical system that accepts motion inputs and yields force outputs is defined as an impedance [11].

²A physical system that accepts force inputs and yields motion outputs is defined as an admittance [11].

³Advanced Telecommunications Research Institute International - Japan

⁴www.sarcos.com

⁵Product of SensAble Technologies, Inc., www.sensable.com

tasks ranging from accurate motion in free-space to stable dynamic interaction with rigid environments. It is possible to improve the performance of both control algorithms through specific hardware modifications but such modifications result in a predisposition to Impedance Control or Admittance Control since their hardware requirements are different. To have no predisposition and instead have complete flexibility in choosing the “best” controller for any given task, we propose a control strategy that unifies Impedance Control and Admittance Control based on the concept of switched systems [17]. Our approach results in a continuous spectrum of hybrid systems with the systems under Impedance Control and Admittance Control at two ends of the spectrum. Note that this strategy is different from improving the performance of either the Impedance Controller or the Admittance Controller, which has been the focus of previous works. It is needless to say that our hybrid system approach can potentially benefit from improved versions of impedance control and admittance control. The current paper is an extension of the conference paper [23], where the basic idea of the hybrid controller was presented. Compared to [23] the current paper contains an extended discussion, an improved case study, and an outlook on the multi-body case.

We do not discuss the procedure for choosing the “best” controller for a given task and environment but derive conditions for stability of dynamic interaction and stability of the switched system in which controllers may be continually switched to deal with changing environments. Through numerical simulations and experimental results it is shown that our hybrid control design improves stability and performance characteristics of interaction controllers which have been restricted to Impedance Control and Admittance Control thus far.

This paper is organized as follows. In Section 2 we present background material on Impedance and Admittance Control implementations and highlight their differences and limitations through numerical simulations. To improve the performance of dynamic interaction, we propose a hybrid control strategy in Section 3. The closed-loop system dynamics is analyzed and conditions for stable behavior are obtained. The stability and performance characteristics of the hybrid controller is demonstrated through simulations in Section 4 and experiments in Section 5. Concluding remarks are provided in Section 6.

2 Background

2.1 Control Objective

Consider a single degree-of-freedom system in which a mass interacts with an environment. Let m and x be the generalized inertia and displacement of the mass, respectively, and let F and F_{ext} be the control force and external force of the environment acting on the mass. Both F and F_{ext} are measured positive in the direction of positive displacement. The equation of motion of the mass can be written as follows

$$m\ddot{x} = F + F_{ext} \quad (1)$$

The control objective for both Impedance and Admittance Control is to design the control force F that will establish the following second-order relationship between F_{ext} and e

$$M_d \ddot{e} + D_d \dot{e} + K_d e = F_{ext} \quad (2)$$

where e is the deviation of x from some desired equilibrium trajectory x_0

$$e = (x - x_0) \quad (3)$$

and M_d , D_d and K_d are positive constants that represent the desired inertia, damping and stiffness, respectively. The transfer function between e and F_{ext} is denoted by

$$G_d(s) = \frac{1}{M_d s^2 + D_d s + K_d} \quad (4)$$

2.2 Impedance Control

In Impedance Control, the controller is a mechanical impedance and consequently the controlled plant is treated as a mechanical admittance. If we assume an ideal force-controlled system, i.e., a system which allows us to command a desired force of interaction with the environment F_d , the relationship in Eq. 2 can be achieved by the control system in Fig. 1 with $F_d = F_{ext}$.

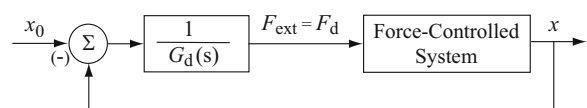


Fig. 1 The concept of Impedance Control, assuming an ideal force-controlled system

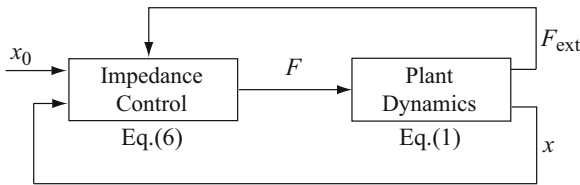


Fig. 2 Implementation of Impedance Control

The control force F for Impedance Control can be formulated from Eqs. 1 and 2 as follows

$$\begin{aligned} F &= m\ddot{x} - F_{\text{ext}} \\ &= (m - M_d)\ddot{x} + (M_d\ddot{x}_0 - D_d\dot{e} - K_de) \end{aligned} \quad (5)$$

Since acceleration feedback is not desirable, we substitute $\ddot{x}$ from Eq. 1 into Eq. 5 to get the final Impedance Control law

$$F = \left(\frac{m}{M_d} - 1 \right) F_{\text{ext}} + m\ddot{x}_0 - \frac{m}{M_d} (D_d\dot{e} + K_de) \quad (6)$$

which now includes a feedback of the force of interaction with the environment instead of the acceleration (see Fig. 2). In the ideal case, where modeling errors and measurement uncertainty is absent, the control law in Eq. 6 exactly implements the desired behavior in Eq. 2 for the model in Eq. 1.

2.3 Admittance Control

In Admittance Control, the plant is position-controlled and not force-controlled. The position-controlled plant behaves as a mechanical impedance and hence the controller is designed to be a mechanical admittance. Let us consider a position-controlled system in which the motion of the mass follows a given desired trajectory x_d without any tracking error, i.e., $x = x_d$. Then, the impedance relationship in Eq. 2 can be implemented by the control system in Fig. 3.

The position-controlled system is comprised of the plant dynamics and the position controller, as shown in Fig. 4. The position controller can be implemented

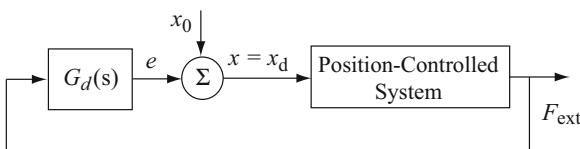


Fig. 3 The concept of Admittance Control, assuming an ideal position-controlled system

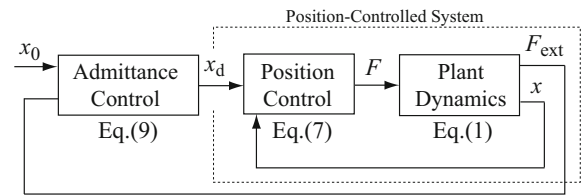


Fig. 4 Implementation of Admittance Control

using a PD regulation controller (see [16], for example) of the form

$$F = k_p (x_d - x) - k_v \dot{x} \quad (7)$$

with positive gains k_p and k_v . Substituting Eq. 7 into Eq. 1 and rewriting Eq. 2 after replacing x with x_d , the complete system dynamics can be written as follows

$$m\ddot{x} + k_v \dot{x} + k_p (x - x_d) = F_{\text{ext}} \quad (8)$$

$$\begin{aligned} M_d(\ddot{x}_d - \ddot{x}_0) + D_d(\dot{x}_d - \dot{x}_0) \\ + K_d(x_d - x_0) = F_{\text{ext}} \end{aligned} \quad (9)$$

It should be noted that the impedance controller in Eq. 6 uses static state feedback whereas the admittance control law in Eqs. 7 and 9 uses dynamic feedback based on the two additional states x_d and $\dot{x}_d$.

The following section provides a simulation-based comparison of Impedance and Admittance Control. It should be mentioned that the performance of Admittance Control depends on the particular choice of the underlying position controller. In our simulation based comparison, we utilize the controller given by Eq. 7, while other choices would be possible as well.

2.4 Comparison of Impedance and Admittance Control

Impedance Control and Admittance Control are based on the complementary assumptions of force-controlled system and position-controlled system and as a consequence their stability and performance characteristics are different.

Impedance Control has two main limitations. First, for implementing a stiff desired behavior (large value of K_d) and/or for large inertia re-scaling (large value of m/M_d), the controller gains of the two outer loops in Fig. 2 become large. This leads to amplification of noise that can result in instability. Second, the performance of the system in terms of position accuracy depends on back drivability and amount of friction.

For a system with considerable uncompensated friction, the position accuracy depends on the desired stiffness and damping and as a direct consequence Impedance Control with low desired stiffness often leads to poor position accuracy in free motion [29]. In light of these drawbacks, it is accurate to state that Impedance Control is particularly suitable for implementing compliant behavior in a low friction device. Impedance Control is also robust to uncertainties in model parameters and leads to stable contact even for very stiff environments. In case of $M_d = m$, this can be attributed to passivity of the system with respect to the pair $(\dot{e}, F_{\text{ext}})$. The effects of unmodeled dynamics of robots with force and Impedance Control have been analyzed by several authors, including [2, 8, 22].

The performance and stability characteristics of Admittance Control largely depends on the quality of the underlying position controller [24] which compensates for the effects of unmodeled friction effectively [19, 29]. Admittance Control can therefore be implemented for systems which are not back drivable but its main limitation is instability in contact with stiff environments. This can be observed even for an ideal system without modeling uncertainties or time delay [16]. In contrast to Impedance Control, Admittance Control inherits low control gains for a desired behavior which is stiff and high control gains for a desired behavior which is compliant. This can be verified from Figs. 2 and 4 and the definition of the transfer function in Eq. 4. As a consequence, Admittance Control is better suited to implementing stiff behavior than compliant behavior. Several researchers have successfully implemented Admittance Control in industrial robot manipulators [5, 9, 31] and adaptive control methods [26, 28] have been proposed for dealing with environments with unknown compliance.

2.5 Simulation Example

To illustrate the advantages and disadvantages of Impedance and Admittance Control, we investigate the behavior of the single degree-of-freedom system in Fig. 5. In this figure, the environment is modeled as a linear spring of stiffness k_e . The equation of motion of the mass is given by the relation

$$m\ddot{x} = F + F_{\text{ext}} + F_f \quad (10)$$

where

$$F_{\text{ext}} = -k_e(x - x_0) \quad (11)$$

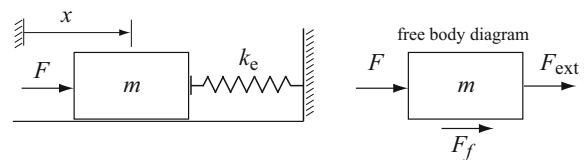


Fig. 5 A single degree-of-freedom system interacting with an environment

is the external force applied by the environment on the mass and F_f is an unmodeled friction term resulting from a non-ideal drive system. This friction term is assumed to have the form

$$F_f = -\text{sign}(\dot{x})(c_v|\dot{x}| + F_c) \quad (12)$$

where c_v and F_c are the coefficients of viscous and Coulomb friction.

Impedance Control was implemented using Eqs. 6 and 10 and Admittance Control was implemented using Eqs. 8, 9 and 10. Modeling uncertainty was introduced in both implementations by replacing the mass m in Eqs. 6 and 8 with an estimated mass $\hat{m}$. It is important to note that two state variables $(x, \dot{x})$ describe the closed-loop system with Impedance Control whereas four state variables $(x, \dot{x}, x_d, \dot{x}_d)$ are required with Admittance Control. This is because of the inner position-control loop in Admittance Control. The parameter values used in our simulations are as follows:

$$\begin{aligned} m &= 1.0 \text{ kg}, \quad \hat{m} = 0.8 \text{ kg} \\ c_v &= 1.0 \text{ Ns/m}, \quad F_c = 3.0 \text{ N} \\ k_p &= 10^6 \text{ N/m}, \quad k_v = 2 \times 0.7\sqrt{k_p m} \text{ Ns/m} \\ M_d &= 1.0 \text{ kg}, \quad K_d = 100 \text{ N/m}, \\ D_d &= 2 \times 0.7\sqrt{K_d M_d} \text{ Ns/m} \end{aligned}$$

The position (PD) controller is designed with high gains, which is common practice, but modeling uncertainty of m is not considered since we can realistically assume that the position control loop is tuned independently. The damping gains are selected such that the behavior is well-damped in the contact-free case. The simulations were based on a sampling time of $T = 1$ ms. It was assumed that feedback of external forces is affected by a Gaussian noise signal (with zero mean and unity variance) and an unmodeled time-delay of $T_d = 2$ ms.

We highlight the performance and stability properties of the closed-loop system with Impedance Control and Admittance Control by simulating a step change

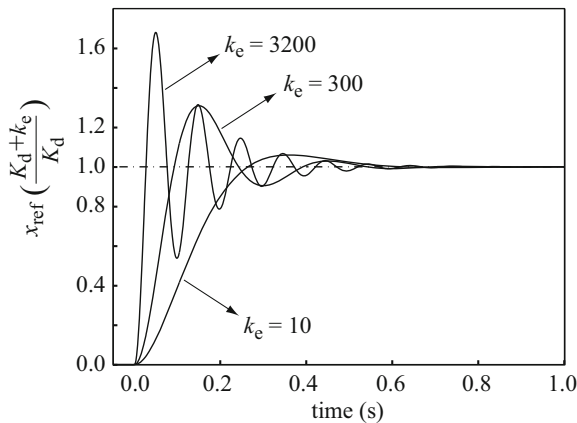


Fig. 6 Ideal trajectories of the mass in Fig. 5 for a step change in the virtual equilibrium position for a soft, an intermediate and a stiff environment

in the virtual equilibrium position x_0 from zero to 1 m (while setting $\ddot{x}_0 = \dot{x}_0 = 0$) at $t = 0$ for a soft, an intermediate, and a stiff environment. We assume that the mass initially is already in contact with the environment such that contact transitions do not appear. To simulate the soft, intermediate and stiff environments, k_e was chosen as 10 N/m, 300 N/m and 3200 N/m, respectively.

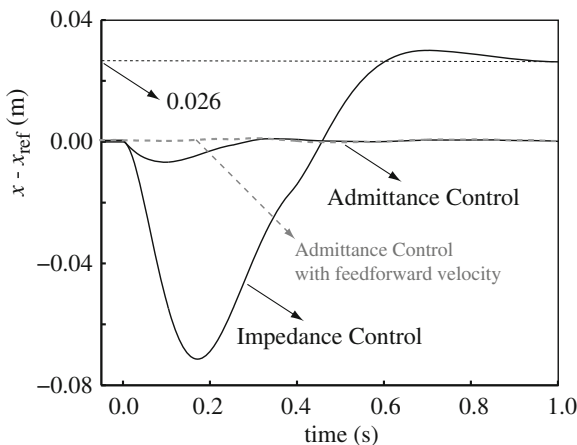


Fig. 7 Deviation in trajectory of mass from its ideal trajectory for Impedance Control and Admittance Control for the soft environment $k_e = 10$ N/m

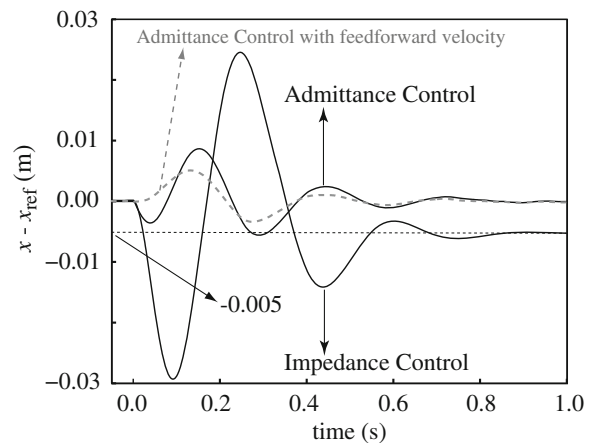


Fig. 8 Deviation in trajectory of mass from its ideal trajectory for Impedance Control and Admittance Control for environment stiffness $k_e = 300$ N/m

The ideal behavior of the closed-loop system, denoted by x_{ref} , is obtained from Eq. 2

$$M_d \ddot{e} + D_d \dot{e} + K_d e = -k_e x$$

$$\Rightarrow M_d \ddot{x}_{\text{ref}} + D_d \dot{x}_{\text{ref}} + (K_d + k_e)x_{\text{ref}} = K_d x_0 \quad (13)$$

and is plotted in Fig. 6. Since different values of k_e result in different final values of x_{ref} , we plot the

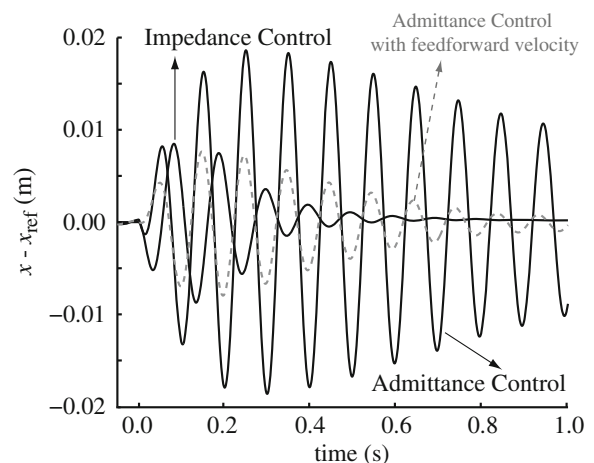


Fig. 9 Deviation in trajectory of mass from its ideal trajectory for Impedance Control and Admittance Control for the stiff environment $k_e = 3200$ N/m

normalized value of x_{ref} in Fig. 6. Figure 7 shows the deviation in the trajectory of the mass from its ideal trajectory for Impedance Control and Admittance Control for the soft environment ($k_e = 10$ N/m). It can be seen that Admittance Control results in good performance whereas Impedance Control results in tracking errors and a steady state error due to uncompensated friction. The results for the environment with intermediate stiffness ($k_e = 300$ N/m) are shown in Fig. 8. By comparing these results with those in Fig. 7 it is clear that Impedance Control results in improved performance with reduced steady state error but the performance of the system deteriorates with Admittance Control. It deteriorates further for the stiff environment ($k_e = 3200$ N/m), which is evident from the large oscillations in Fig. 9. This can be attributed to the high gains of the underlying position (PD) controller and the time-delay of force feedback. For the stiff environment, Impedance Control however provides very good performance with negligible steady state error.

The performance of Admittance Control also depends on the choice of the inner loop position controller in Eq. 7. As a comparison, the dashed lines in Figs. 7 to 9 additionally show the results with the modified position controller $F = k_p(x_d - x) + k_v(\dot{x}_d - \dot{x})$ that includes a feedforward term in the desired velocity. This controller performs better than the regulation controller from Eq. 7, but one can observe the same trend for its stability properties.

3 Hybrid System Framework

3.1 Motivation

Impedance Control provides very good performance when the environment is stiff but results in poor accuracy when the environment is soft. In contrast, Admittance Control provides very good performance for soft environments but results in contact instability for stiff environments. The complementary characteristics of the two controllers is well known (see [32], for example) and is qualitatively illustrated here in Fig. 10. Both Impedance Control and Admittance Control result in unsatisfactory performance for intermediate values of environment stiffness and are unsuitable

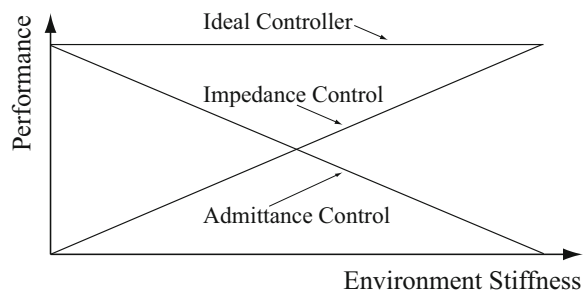


Fig. 10 Qualitative illustration of the performance of Impedance Control and Admittance Control for different environment stiffness

when there are large changes in the stiffness. This limitation can be attributed to their fixed causality. An ideal controller should provide consistently good performance, independent of the environment stiffness. In the next section we propose a control strategy that overcomes the limitations of fixed-causality controllers by continually switching between impedance and admittance causality.

3.2 Framework

For the single degree-of-freedom system described by Eq. 1, we propose to switch the controller between impedance and admittance causality as follows

$$F = \begin{cases} F_1 & : t_0 + k\delta \leq t \leq t_0 + (k+1-n)\delta \\ F_2 & : t_0 + (k+1-n)\delta < t < t_0 + (k+1)\delta \end{cases} \quad (14)$$

where t_0 is the initial time, δ is the switching period, $n \in [0, 1]$ is the duty cycle, k is an integer that takes on values $0, 1, \dots$, F_1 is the static state feedback law given by Eq. 6, and F_2 is the dynamic controller described by Eqs. 7 and 9. This is explained with the help of Fig. 11.

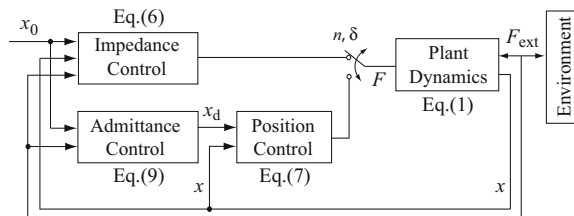


Fig. 11 Controller switching between impedance and admittance causality

With Impedance Control, the closed-loop system behavior is described by Eq. 2. This can be verified by substituting Eq. 6 into Eq. 1. With Admittance Control, the closed-loop system behavior is described by Eqs. 8 and 9. If the environment is modeled as a linear spring

$$F_{\text{ext}} = -k_e(x - x_0) = -k_e e \quad (15)$$

and the virtual equilibrium position is assumed fixed, i.e.,

$$\dot{x}_0 = \ddot{x}_0 = 0 \quad (16)$$

the hybrid system has the following descriptions

$$\begin{aligned} \dot{X}_i &= A_i X_i : t_0 + k\delta \leq t \leq t_0 + (k+1-n)\delta \\ \dot{X}_a &= A_a X_a : t_0 + (k+1-n)\delta < t < t_0 \\ &\quad + (k+1)\delta \end{aligned} \quad (17)$$

where $X_i = (e \ \dot{e})^T$, $X_a = (e \ \dot{e} \ e_d \ \dot{e}_d)^T$, $e_d = (x_d - x_0)$ and

$$\begin{aligned} A_i &= \begin{bmatrix} 0 & 1 \\ -(K_d + k_e)/M_d & -D_d/M_d \end{bmatrix} \\ A_a &= \begin{bmatrix} 0 & 1 & 0 & 0 \\ -(k_p + k_e)/m & -k_v/m & k_p/m & 0 \\ 0 & 0 & 0 & 1 \\ -k_e/M_d & 0 & -K_d/M_d & -D_d/M_d \end{bmatrix} \end{aligned}$$

When the system switches from Impedance to Admittance Control, two additional states are introduced. These states, e_d and $\dot{e}_d$, are conveniently chosen to maintain continuity in the control force F and its derivative. Equation 7, which gives the expression for the control force in Admittance Control, can be written as follows

$$\begin{aligned} x_d &= x + \frac{1}{k_p}(F + k_v \dot{x}) \\ \Rightarrow e_d &= e + \frac{1}{k_p}(F + k_v \dot{e}) \\ \dot{x}_d &= \dot{x} + \frac{1}{k_p}(\dot{F} + k_v \ddot{x}) \\ &= \dot{x} + \frac{1}{k_p} \left[\dot{F} + \frac{k_v}{m}(F + F_{\text{ext}}) \right] \\ \Rightarrow \dot{e}_d &= \dot{e} + \frac{1}{k_p} \left[\dot{F} + \frac{k_v}{m}(F + F_{\text{ext}}) \right] \end{aligned} \quad (18)$$

Substituting the expression for the control force for Impedance Control from Eq. 6 in the above equation it is possible to obtain an expression of the form

$$X_a = S_{ai} X_i, \quad S_{ai} = \begin{bmatrix} I \\ S \end{bmatrix} \quad (19)$$

where I is the identity matrix and entries of $S = [s_{ij}]_{2 \times 2}$ have the expressions

$$\begin{aligned} s_{11} &= 1 - \frac{k_e}{k_p} \left(\frac{m}{M_d} - 1 \right) - \frac{K_d}{k_p} \frac{m}{M_d} \\ s_{12} &= \frac{k_v}{k_p} - \frac{D_d}{k_p} \frac{m}{M_d} \\ s_{21} &= -\frac{m}{M_d} \frac{(K_d + k_e)}{k_p} \left(\frac{k_v}{m} - \frac{D_d}{M_d} \right) \\ s_{22} &= 1 - \frac{k_e}{k_p} \left(\frac{m}{M_d} - 1 \right) - \frac{D_d}{M_d} \left(\frac{k_v}{k_p} - \frac{D_d}{k_p} \frac{m}{M_d} \right) \\ &\quad - \frac{K_d}{k_p} \frac{m}{M_d} \end{aligned} \quad (20)$$

When the system switches from Admittance to Impedance Control, the state variable mapping is given by the relation

$$X_i = S_{ia} X_a, \quad S_{ia} = [I \ O] \quad (21)$$

where O is the 2×2 matrix with zero entries.

It should be mentioned that for $n = 1$ the hybrid system uses admittance control with periodic resetting. This specific hybrid system is similar to the closed loop system in [32], where reinitialization of the desired trajectory is performed periodically.

3.3 Stability Analysis

The stability analysis presented here is very similar to the stability analysis of the hybrid system with reversible transducers [7], investigated earlier by one of the authors. Knowing the states of the system at $t = t_0 + k\delta$, the states at time $t = t_0 + (k+1)\delta$, $k = 0, 1, 2, \dots$ can be obtained using Eqs. 17, 19 and 21 as follows

$$\begin{aligned} X_i(t_0 + (k+1-n)\delta) &= e^{A_i(1-n)\delta} X_i(t_0 + k\delta) \\ X_a(t_0 + (k+1-n)\delta) &= S_{ai} X_i(t_0 + (k+1-n)\delta) \\ X_a(t_0 + (k+1)\delta) &= e^{A_a n\delta} \times \\ &\quad X_a(t_0 + (k+1-n)\delta) \\ X_i(t_0 + (k+1)\delta) &= S_{ia} X_a(t_0 + (k+1)\delta) \end{aligned} \quad (22)$$

$$\Rightarrow X_i(t_0 + (k+1)\delta) = S_{ia} e^{A_a n\delta} S_{ai} e^{A_i(1-n)\delta} \times X_i(t_0 + k\delta) \quad (23)$$

We now define *Discrete Equivalent Subsystem* based on the definition of *Discrete Equivalent* in Das and Mukherjee [7].

Definition 1 *Discrete Equivalent Subsystem (DES)*: The time-invariant linear system

$$\dot{X} = A_{eq}X \quad (24)$$

is a DES of a switched linear system if state variables of the DES assume identical values of a subset of the states of the switched system at regular intervals of time, starting from the same initial condition.

Based on the above definition, the system described by Eq. 24 is a DES of the switched system described by Eqs. 17, 19 and 21 with

$$A_{eq} = \frac{1}{\delta} \ln[S_{ia} e^{A_a n \delta} S_{ai} e^{A_i (1-n)\delta}] \quad (25)$$

Since A_{eq} is the logarithm of a matrix, the issues of existence and uniqueness arise. We address these issues after we present the condition for stability of the switched system.

Theorem 1 *Exponential Stability*:⁶ The equilibrium $X_i = 0$ of the switched system described by Eqs. 17, 19 and 21 is exponentially stable if and only if A_{eq} of the DES in Eq. 25 is Hurwitz.

Proof We prove sufficiency first. For convenience, we define the following norms:

$$\begin{aligned} \eta_1 &= \|A_i\| & c_1 &= \|S_{ai}\| \\ \eta_2 &= \|A_a\| & c_2 &= \|S_{ia}\| \end{aligned} \quad (26)$$

For the DES, we assume $\|X(t_0)\| = \epsilon$. Since, A_{eq} is Hurwitz, we have

$$\begin{aligned} X(t) &= e^{A_{eq}(t-t_0)} X(t_0) \\ \Rightarrow \|X(t)\| &\leq \|e^{A_{eq}(t-t_0)}\| \|X(t_0)\| \\ &\leq \gamma e^{-\lambda(t-t_0)} \epsilon \end{aligned} \quad (27)$$

where $\gamma, \lambda > 0$ are positive numbers. Since the states of the switched system and the DES assume identical values at $t = t_0 + k\delta, k = 0, 1, 2, \dots$, the states of the switched system satisfy

$$\|X_i(t_0 + k\delta)\| \leq \gamma e^{-\lambda k\delta} \quad (28)$$

Now consider the time interval $t_0 + k\delta \leq t \leq t_0 + (k+1)\delta$. Within this interval, first consider the subinterval

$(t_0 + k\delta) \leq t \leq (t_0 + k\delta + (1-n)\delta)$ where Impedance Control is used. Using Eq. 28 and the relations $t = t_0 + k\delta + \tau_1, 0 \leq \tau_1 \leq (1-n)\delta$, we have

$$\begin{aligned} X_i(t_0 + k\delta + \tau_1) &= e^{A_i \tau_1} X_i(t_0 + k\delta) \\ \Rightarrow \|X_i(t_0 + k\delta + \tau_1)\| &\leq \|e^{A_i \tau_1}\| \|X_i(t_0 + k\delta)\| \\ &\leq e^{\eta_1 \tau_1} \gamma e^{-\lambda k\delta} \\ &\leq \gamma e^{(\eta_1 + \lambda)\tau_1} e^{-\lambda(k\delta + \tau_1)} \\ \Rightarrow \|X_i(t)\| &\leq \gamma e^{(\eta_1 + \lambda)(1-n)\delta} e^{-\lambda(t-t_0)} \end{aligned} \quad (29)$$

From Eq. 29 we directly get

$$\|X_i(t_0 + k\delta + (1-n)\delta)\| \leq \gamma e^{\eta_1(1-n)\delta} e^{-\lambda k\delta} \quad (30)$$

At $t = t_0 + k\delta + (1-n)\delta$, Admittance Control is invoked and X_a defines the new states of the system. Using Eq. 30 and the second relation in Eq. 22 we get

$$\begin{aligned} \|X_a(t_0 + k\delta + (1-n)\delta)\| &\leq \|S_{ai}\| \gamma e^{\eta_1(1-n)\delta} e^{-\lambda k\delta} \\ &= c_1 \gamma e^{\eta_1(1-n)\delta} e^{-\lambda k\delta} \end{aligned} \quad (31)$$

Now consider the subinterval of time $(t_0 + k\delta + (1-n)\delta) < t < (t_0 + (k+1)\delta)$ where Admittance Control is used. For $t = t_0 + k\delta + (1-n)\delta + \tau_2, 0 < \tau_2 < n\delta$, we have

$$\begin{aligned} X_a(t_0 + k\delta + (1-n)\delta + \tau_2) \\ = e^{A_a \tau_2} X_a(t_0 + k\delta + (1-n)\delta) \end{aligned}$$

Using Eq. 31 we get

$$\begin{aligned} \|X_a(t_0 + k\delta + (1-n)\delta + \tau_2)\| \\ \leq \|e^{A_a \tau_2}\| \|X_a(t_0 + k\delta + (1-n)\delta)\| \\ \leq e^{\eta_2 \tau_2} c_1 \gamma e^{\eta_1(1-n)\delta} e^{-\lambda k\delta} \end{aligned}$$

Using Eq. 21 we can write

$$\begin{aligned} \|X_i(t_0 + k\delta + (1-n)\delta + \tau_2)\| \\ \leq \|S_{ia}\| \|X_a(t_0 + k\delta + (1-n)\delta + \tau_2)\| \\ \leq c_2 e^{\eta_2 \tau_2} c_1 \gamma e^{\eta_1(1-n)\delta} e^{-\lambda k\delta} \\ \leq c_1 c_2 \gamma e^{(\eta_1 + \lambda)(1-n)\delta} e^{(\eta_2 + \lambda)n\delta} e^{-\lambda(t-t_0)} \end{aligned} \quad (32)$$

From Eqs. 29 and 32 we deduce

$$\begin{aligned} \|X_i(t)\| &\leq \kappa e^{-\lambda(t-t_0)}, \\ \kappa &\equiv c_1 c_2 \gamma e^{(\eta_1 + \lambda)(1-n)\delta} e^{(\eta_2 + \lambda)n\delta} \end{aligned} \quad (33)$$

for $t \in [t_0 + k\delta, t_0 + (k+1)\delta], k = 0, 1, 2, \dots$, which implies exponential stability of $X_i = 0$.

Since the switched system and the DES have the same states at regular intervals of time, the proof of necessity simply follows from the fact that the DES is not exponentially stable if A_{eq} is not Hurwitz. $\square$

⁶A slightly different version of this theorem and proof appears in the paper by Das and Mukherjee [7].

Remark 1 The necessary and sufficient condition for A_{eq} in Eq. 25 to exist is that the matrix $M = S_{ia} e^{A_a n \delta} S_{ai} e^{A_i (1-n)\delta}$ is non-singular [10]. The uniqueness of A_{eq} is not an issue since all solutions of A_{eq} obtained from Eq. 25 will have the same eigenvalues [10], and the results of Theorem 1 are based on the Hurwitz property of A_{eq} and not on the entries of A_{eq} .

Remark 2 Our stability analysis was based on switching from Impedance Control to Admittance Control and back to Impedance Control. This sequence leads to a DES defined by the logarithm of the 2×2 matrix $S_{ia} e^{A_a n \delta} S_{ai} e^{A_i (1-n)\delta}$, shown in Eq. 25. If we switched from Admittance Control to Impedance Control and back to Admittance Control, the DES would be defined by the logarithm of the 4×4 matrix $S_{ai} e^{A_i (1-n)\delta} S_{ia} e^{A_a n \delta}$ instead. This does not change the stability analysis, since for any two matrices $A \in R^{m \times n}$ and $B \in R^{n \times m}$ with $m < n$, the eigenvalues of AB and BA are the same except for the $(n - m)$ additional eigenvalues of BA which are identically zero [14].

Remark 3 The control framework and its subsequent stability analysis is based on a specific implementation of the position controller within the Admittance Control law. Therefore, strictly speaking, the analysis is valid for this position controller only. However, it is possible to perform the same analysis with a different position controller following the steps given in the previous sections. In any case, the stability analysis in this paper requires a linear closed loop dynamics of Impedance and Admittance Control.

3.4 Extension to Multi-DOF Systems

The numerical stability analysis of the previous section is limited to a linear hybrid closed loop dynamics and can be applied mutatis mutandis to higher dimensional linear systems. By utilizing an inverse dynamics based position controller in the implementation of Admittance Control, one thus can apply the same approach to a multi-body system. Let q denote the vector of generalized coordinates and τ the vector of generalized forces (control inputs). Then, the robot dynamics can be written as

$$M(q)\ddot{q} + h(q, \dot{q}) = \tau + \tau_{ext}$$

where $M(q)$ is the inertia matrix and the vector $h(q, \dot{q})$ contains the generalized gravitational, centrifugal and Coriolis forces. The external torques τ_{ext} are assumed to be available from measurement. For a desired impedance

$$M_d \ddot{e} + D_d \dot{e} + K_d e = \tau_{ext}$$

with $e = (q - q_0)$ and positive definite constant matrices M_d, D_d, K_d , the conventional impedance control law is given by

$$\tau = h(q, \dot{q}) + \left(M(q) M_d^{-1} - I \right) \tau_{ext} + M(q) M_d^{-1} (M_d \ddot{q}_0 - D_d \dot{e} - K_d e) \quad (34)$$

For the admittance controller, we assume a computed torque based regulation controller. A closed-loop system behavior similar to the one-dimensional case is achieved via the control law

$$\tau = h(q, \dot{q}) + M(q) (K_p(q_d - q) - K_v \dot{q}) - (I - M(q)) \tau_{ext}, \quad (35)$$

with symmetric and positive definite controller gain matrices K_p and K_v , and the outer loop admittance control law

$$M_d(\ddot{q}_d - \ddot{q}_0) + D_d(\dot{q}_d - \dot{q}_0) + K_d(q_d - q_0) = \tau_{ext}. \quad (36)$$

This leads to the closed loop dynamics

$$\ddot{q} + K_v \dot{q} + K_p(q - q_d) = \tau_{ext} \quad (37)$$

$$M_d(\ddot{q}_d - \ddot{q}_0) + D_d(\dot{q}_d - \dot{q}_0) + K_d(q_d - q_0) = \tau_{ext} \quad (38)$$

When switching from impedance to admittance control, the additional state variables q_d and $\dot{q}_d$ of the admittance controller have to be initialized. One possible way for deciding an appropriate switching law is the requirement that the control input should be continuous at the time of switching, which can be achieved via

$$q_d = q + K_p^{-1} \left(K_v \dot{q} - \tau_{ext} + M_d^{-1} A(e, \dot{e}, \tau_{ext}) \right), \\ A(e, \dot{e}, \tau_{ext}) = M_d \ddot{q}_0 - D_d \dot{e} - K_d e + \tau_{ext}.$$

A numerical stability analysis of the hybrid multi-dof system can be carried out in the same manner as that in the previous section. The concept of using a hybrid systems approach for combining Impedance Control and Admittance Control is not necessarily restricted to a single-dof system. However, it should be noted that the need for a linear closed-loop system dynamics imposes a limitation on the choice of

different position controllers and desired impedance behaviors. The extension of this analysis to nonlinear closed-loop dynamics is still left as an open issue for further research.

4 Hybrid Impedance and Admittance Control: A Case Study

4.1 Stability of Nominal Plant

We investigate stability of the single degree-of-freedom system described in Section 2.5 in the absence of friction, model uncertainty, and time delay, i.e., $F_f = 0$, $\hat{m} = m$ and $T_d = 0$. The hybrid system framework in Section 3 is implemented using constant values of n and δ . Figure 12 shows the results of our numerical stability analyses based on Theorem 1. The solid lines show stability boundaries for different values of δ as they vary with respect to k_e and n . With the lower left hand side of the solid lines denoting the stable region, it is clear that a smaller value of δ results in a larger region of stability. In Fig. 12, $n = 0$ denotes Impedance Control and $n = 1$ denotes Admittance Control and therefore one would expect the stability curves for different values of δ to all meet at the stability boundary of standard Admittance Control. However, this is not the case since $n = 1$ represents Admittance Control with periodic resetting of states (at δ time intervals) by the mapping $S_{ai}S_{ia}$. The matrices S_{ai} and S_{ia} were defined earlier in Eqs. 19

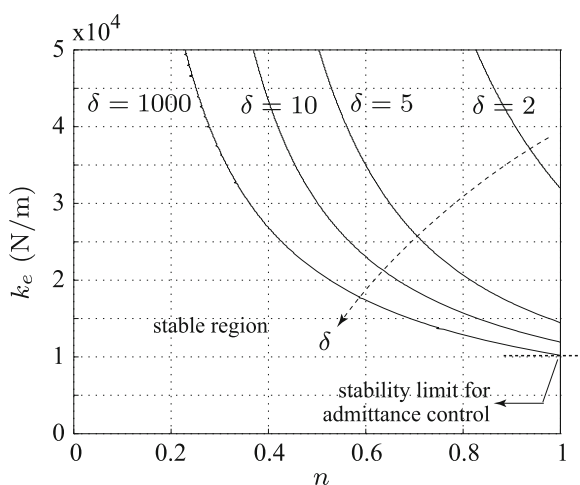


Fig. 12 Stability boundaries for hybrid impedance and admittance control for different values of δ in msec

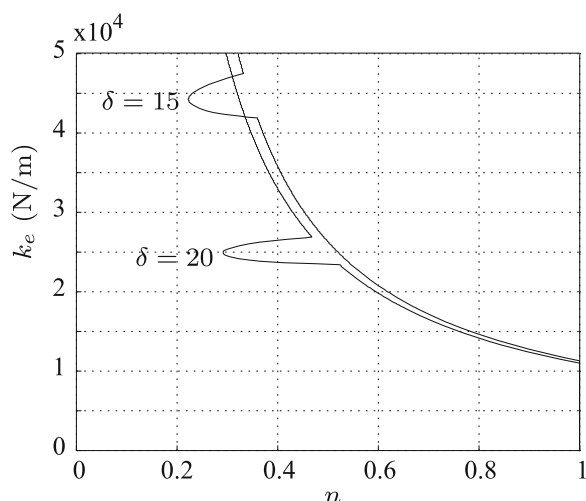


Fig. 13 Stability boundaries for hybrid impedance and admittance control for $\delta = 15$ ms and $\delta = 20$ ms

and 21. For large values of δ , the stability boundary⁷ of the closed-loop system with Admittance Control and periodic resetting ($n = 1$) approaches that of the system with standard Admittance Control but for small values of δ , periodic resetting alone improves stability. The improvement is even greater for $n < 1$, i.e., for Hybrid Impedance and Admittance Control, which includes periodic resetting of states.

Figure 12 is informative but it does not provide the complete picture of closed-loop system stability. A plot of the stability curves for specific values of δ , $\delta = 15$ msec and $\delta = 20$ msec, shown in Fig. 13 reveals an enlarged unstable region for certain values of k_e . Consider for instance the case $n = 0.4$ and $\delta = 20$ ms. If we increase the environment stiffness from $k_e = 2 \times 10^4$ N/m to $k_e = 2.5 \times 10^4$ N/m, the closed-loop system becomes unstable; but if the stiffness is increased further to $k_e = 3 \times 10^4$ N/m, the system becomes stable. Such a phenomenon can also be observed for other parameter values, including the one shown in Fig. 12. However, for $\delta = 1000$ ms, the increased unstable regions become very small and are hardly visible as small dots on the left hand side of the stability border line in Fig. 12. For smaller values of δ , this effect can only be observed for very large values of the external stiffness. Although this phenomenon has been verified through numerical simulations, it is

⁷As seen from Fig. 12, this is characterized by the maximum value of environment stiffness k_e for which the system remains stable.

not intuitive and requires further investigation. A preliminary analysis motivates us to speculate that this effect emerges from resonance between the eigenfrequency of the closed loop system $[(K_d + k_e)/M_d]^{1/2}$ and some multiple of the switching frequency $2\pi/\delta$.

Remark 4 The hybrid switching procedure followed in this paper is fundamentally different from using averaged control effort $F = (1 - n)F_{\text{imp}} + nF_{\text{adm}}$, where F_{imp} and F_{adm} are the control inputs from Impedance and Admittance Control. Indeed, simulation based on averaged control fails to provide good results. An evaluation for the parameters from the case study in this section showed that the maximum allowed⁸ external stiffness is lower than for the plain admittance controller, except for very small values of n - see Fig. 14. This can be explained by the fact that in case of averaged control the (low gain) impedance controller basically works as a disturbance to the (high gain) admittance controller.

Remark 5 The simulation results of this case study are valid for an Admittance Control law implemented based on the underlying position controller in Eq. 7. As already mentioned in Remark 3, it is possible to repeat the analysis and simulations for other position controllers. For a PD controller using a feedforward term in the desired velocity, i.e., $F = k_p(x_d - x) + k_v(\dot{x}_d - \dot{x})$, the stability analysis is quite similar and the simulation results were found to be quite similar to those reported in this section.

4.2 Performance of Uncertain Plant

We study the performance of the single degree-of-freedom system described in Section 2.5 in the presence of friction, model uncertainty, and time delay. Similar to Section 2.5, we simulate a step change in the virtual equilibrium position x_0 from zero to 1 m (while setting $\ddot{x}_0 = \dot{x}_0 = 0$) for a soft, an intermediate, and a stiff environment. To simulate the soft, intermediate and stiff environments, k_e was chosen as 10 N/m, 300 N/m and 3200 N/m, respectively.

Figure 15 plots the response for a soft environment ($k_e = 10$ N/m) for different values of n and $\delta = 20$ ms. Also shown in the figure are the response for Impedance and Admittance Control, taken from

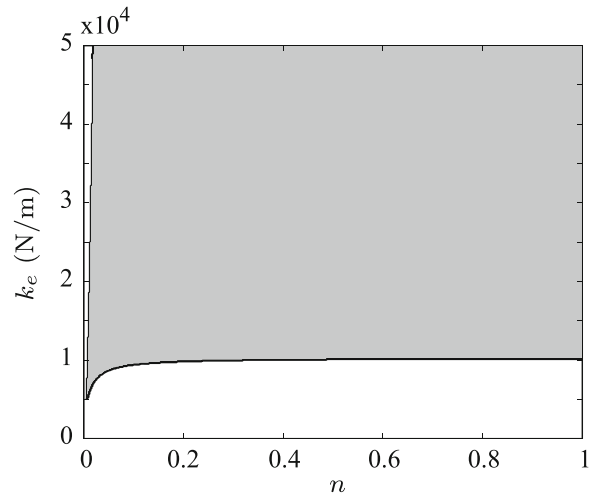


Fig. 14 Stability boundary for averaged hybrid impedance and admittance control - the unstable region is shaded grey

Fig. 7. It can be seen that for small values of n Hybrid Impedance and Admittance Control results in trajectories similar to that of Impedance Control and leads to relatively large steady state error. As the value of n is increased, the steady state error reduces, and in the limit the response tends to that of Admittance Control with zero steady state error. Clearly, we can interpolate between the response of Impedance Control and Admittance Control by properly choosing the value of n .

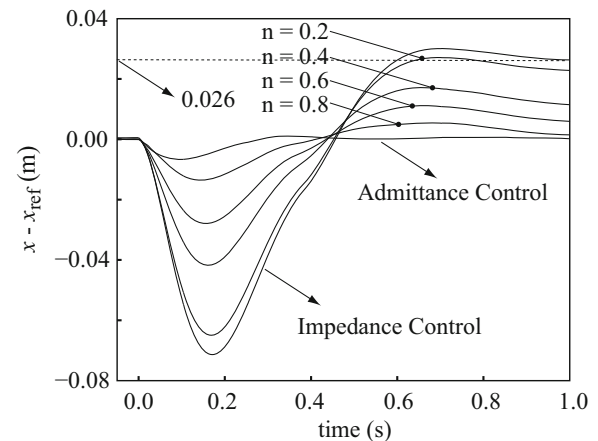


Fig. 15 Deviation in trajectory of mass from its ideal response during interaction with a soft environment ($k_e = 10$ N/m). The different plots are for Impedance Control, Admittance Control, and Hybrid Impedance and Admittance Control with $n = \{0.2, 0.4, 0.6, 0.8\}$

⁸Leading to a stable closed loop behavior

Simulation results for an environment with intermediate stiffness ($k_e = 300$ N/m) are shown in Fig. 16. Also shown in the figure are the response for Impedance and Admittance Control, taken from Fig. 8. The plots confirm the observation made earlier from Fig. 15, namely, the duty cycle (n) can be chosen to obtain a response that is an interpolation of the responses of Impedance Control and Admittance Control. The same observation can be made for a stiff environment as well. The simulation results for a stiff environment, shown in Fig. 17, indicate that low values of n result in small oscillations and higher values of n result in large oscillations. This is in conformity with the results in Fig. 9 where Impedance Control results in a more stable response (less oscillatory) in comparison to Admittance Control. The plots in Fig. 9 were not superimposed on top of the plots in Fig. 17 to avoid clutter.

4.3 Flexibility to Adapt to Changing Environment

In the simulations presented thus far we considered constant values of n and δ . Clearly, if the stiffness of the environment is known or can be observed, the value of n can be adapted to the contact stiffness with the goal of achieving stable motion in stiff contact and good position accuracy in soft contact or in free motion. This is illustrated with the simulation of the single degree-of-freedom mass in Section II-E with

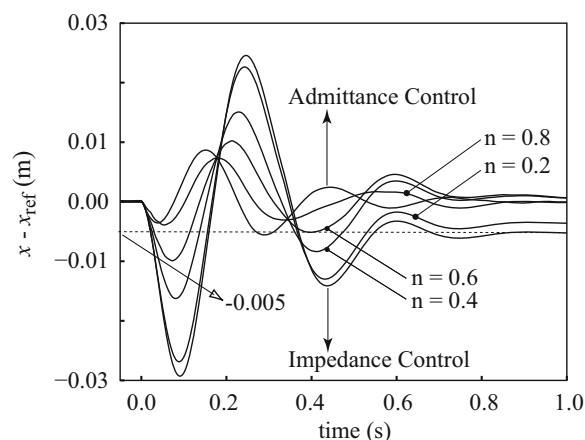


Fig. 16 Deviation in trajectory of mass from its ideal response during interaction with an intermediate environment ($k_e = 300$ N/m). The different plots are for Impedance Control, Admittance Control, and Hybrid Impedance and Admittance Control with $n = \{0.2, 0.4, 0.6, 0.8\}$

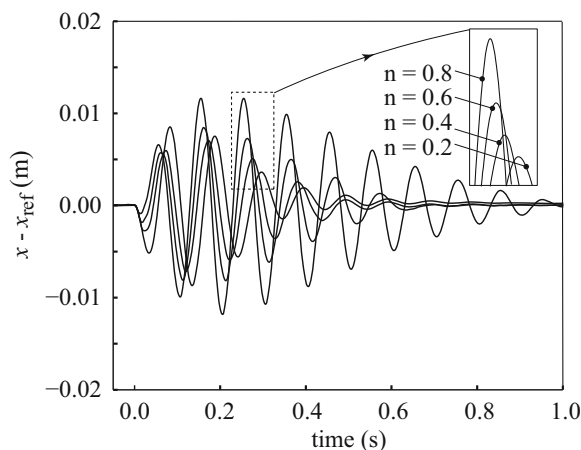


Fig. 17 Deviation in trajectory of mass from its ideal response during interaction with a stiff environment ($k_e = 3200$ N/m). The different plots are for Hybrid Impedance and Admittance Control with $n = \{0.2, 0.4, 0.6, 0.8\}$. The plots with higher oscillations correspond to higher values of n

friction, model uncertainty and time delay. The sampling time was chosen to be $T = 1$ ms and the value of δ was chosen as 20 ms. To switch between impedance and admittance control at the time of sampling only, the values of n were restricted to the set $n \in \{0.00, 0.05, 0.10, \dots, 0.95, 1.00\}$. The contact stiffness k_e was assumed to vary according to the plot in Fig. 18. It can be seen that k_e increases linearly from zero to 1500 N/m, then remains constant for a short duration of time, increases again till it reaches its maximum value of 3000 N/m, remains constant briefly, and finally settles to a small value of 10 N/m. The adaptation in the value of n and the commanded virtual equilibrium position x_0 are also plotted in Fig. 18. We used the simple linear adaptation law

$$n = \begin{cases} 0 & : k_e \geq k_{e,imp} \\ (k_{e,imp} - k_e) / (k_{e,imp} - k_{e,adm}) & : k_{e,adm} < k_e < k_{e,imp} \\ 1 & : k_e \leq k_{e,adm} \end{cases} \quad (39)$$

where $k_{e,imp}$ is the minimum stiffness value for which Impedance Control shall be used, and $k_{e,adm}$ is the maximum stiffness value for which Admittance Control shall be used. The values of $k_{e,imp}$ and $k_{e,adm}$ should be chosen such that the graph of $n(k_e)$ does not intersect the stability borders in Fig. 12 for the chosen value of δ . In this simulation we used $k_{e,imp} = 3000$ N/m and $k_{e,adm} = 0$.

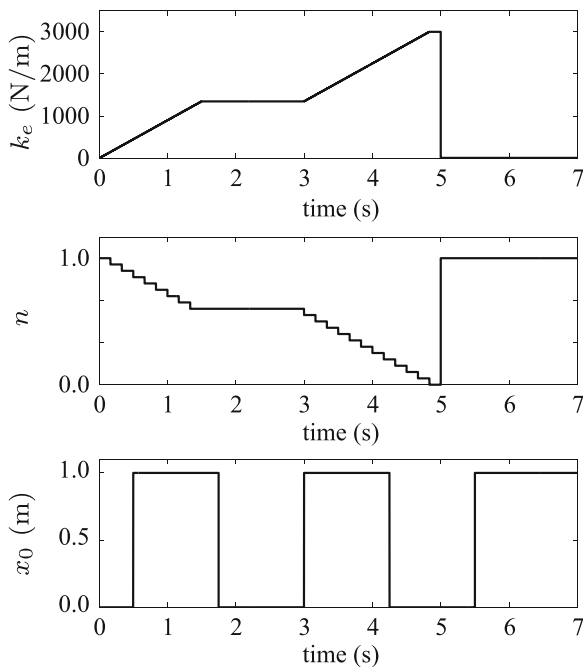


Fig. 18 Time varying contact stiffness k_e , corresponding adaptation in the value of n , and commanded equilibrium position x_0

The ideal behavior of the closed-loop system, x_{ref} , is shown in Fig. 19. Figure 20 shows the deviation of x from the ideal behavior x_{ref} for Impedance Control, Admittance Control, and Hybrid Impedance and Admittance Control. We investigate two cases for the hybrid controller. For the first case, the duty cycle was assumed to be constant, equal to 0.5. For the second case, the duty cycle (n) was adapted according to the plot in Fig. 19. The following observations can be made from the plots in Fig. 20:

- Impedance Control has a considerable steady state error, especially for low contact stiffness (during 6–7 sec).

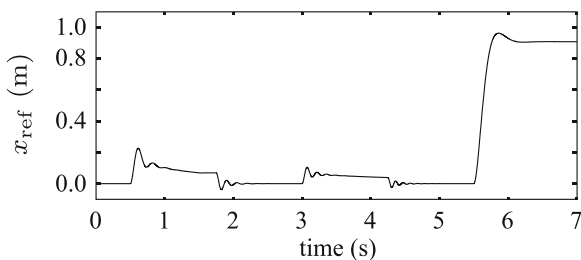


Fig. 19 Ideal motion of the closed-loop system, x_{ref} , for the time varying contact stiffness and commanded equilibrium position shown in Fig. 18

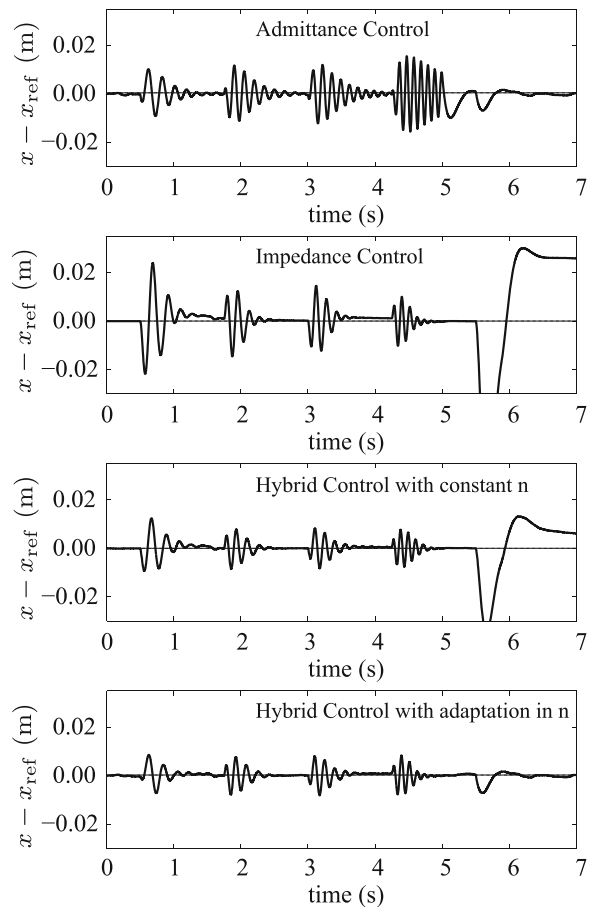


Fig. 20 Comparison of errors ($x - x_{\text{ref}}$) for Admittance Control, Impedance Control, and Hybrid Impedance and Admittance Control with $n = 0.5$ and adaption in the value of n

- Admittance Control shows an oscillatory behavior when the contact stiffness is high (during 4–5 sec, for example) but the steady state error is low at all times.
- Hybrid Impedance and Admittance Control with a constant duty cycle ($n = 0.5$) results in a response that has better steady state accuracy than Impedance Control and less oscillations than Admittance Control. For an intermediate value of stiffness, Hybrid Impedance and Admittance Control results in an error that is smaller than both Impedance Control and Admittance Control - see the response during 1.5–3 sec, for example.
- Through adaptation of the duty cycle (n), it is possible to further reduce the oscillatory response (1.5–3 sec, for example) as well as the steady state error (6–7 sec, for example) as compared to the

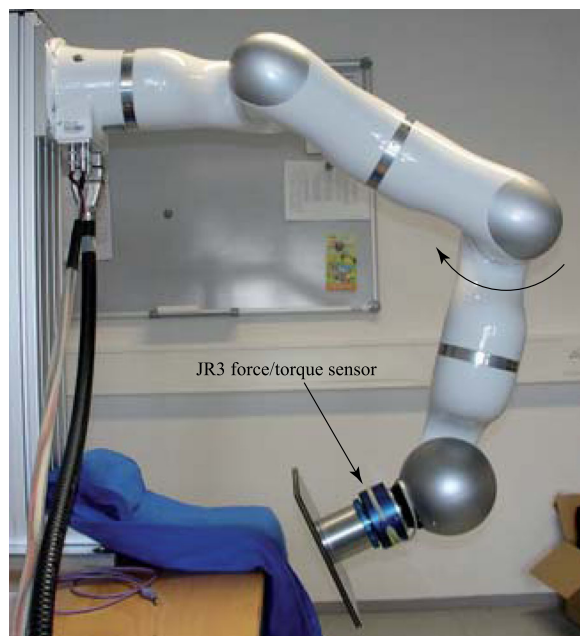


Fig. 21 Experimental setup showing KUKA-DLR lightweight arm

controller with fixed n . This indicates the potential for optimizing the performance of the hybrid controller.

5 Experimental Verification

We compared the performance of Hybrid Impedance and Admittance Control with that of Impedance Control and Admittance Control using the KUKA⁹-DLR lightweight arm, shown in Fig. 21. The arm is equipped with a JR3¹⁰ force/torque sensor at the wrist for measurement of external forces and torques. The elbow joint of the robot (marked in Fig. 21) was actively controlled while all other joints were held fixed. The external torque acting on the elbow joint was computed by mapping the measurements from the force/torque sensor via the transposed Jacobian onto the joint space.

For Impedance Control, the joint torque was computed using Eq. 6 based on measurement of the external torque and position of the joint, F_{ext} and x . For Admittance Control, the joint torque was computed

⁹<http://www.kuka-robotics.com>

¹⁰<http://www.jr3.com>

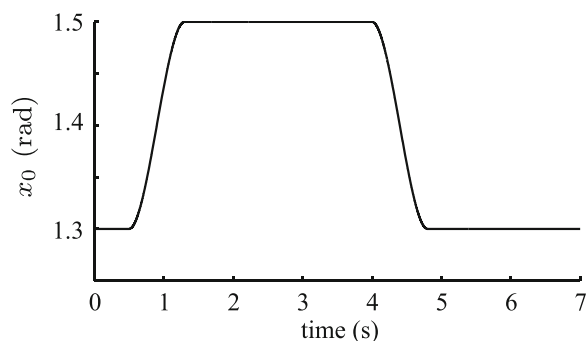


Fig. 22 Commanded virtual equilibrium position of the elbow joint

using Eqs. 7 and 9, also based on measurements of F_{ext} and x . The torque control interface of the arm was used to implement both Impedance Control and Admittance Control with a sampling time of 1 ms. The computed torques were used as reference signals for the robot's high-speed inner-loop torque controller which uses joint torque measurements. The inertia apparent from the elbow joint was computed from CAD data as

$$\hat{m} = 0.433 \text{ kgm}^2 \quad (40)$$

and the desired impedance model and control parameters were chosen as follows

$$k_p = 5000 \text{ Nm/rad}, \quad k_v = 15 \text{ Nms/rad}$$

$$M_d = \hat{m}, \quad K_d = 100 \text{ N/m},$$

$$D_d = 2 \times 0.7 \sqrt{K_d M_d} \text{ Ns/m}$$

In our experiment, the elbow joint was commanded to rotate by 0.2 rad (from 1.3 rad to 1.5 rad) such that the end-effector comes in contact with the supporting table (see Fig. 21) and then rotate back to its original angle. Since the environmental stiffness was unknown,

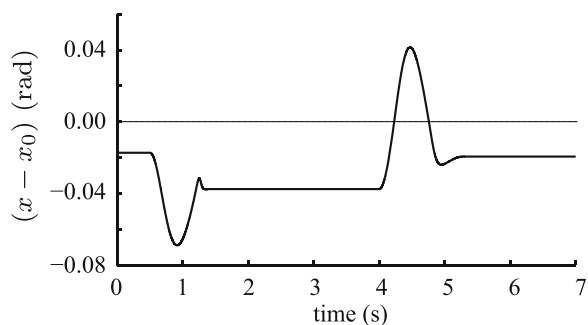


Fig. 23 Experimental result with Impedance Control

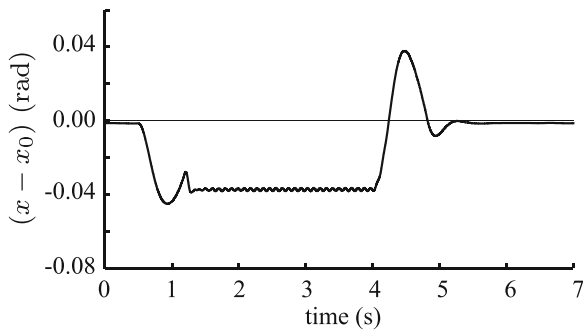


Fig. 24 Experimental result with Admittance Control

it was not possible to determine the ideal trajectory (x_{ref}), and therefore we compared the trajectory generated by Hybrid Impedance and Admittance Control with those generated by Impedance Control and Admittance Control. The commanded virtual equilibrium position x_0 is shown in Figs. 22, 23 and 24 show the errors in the trajectories generated by Impedance Control and Admittance Control. It can be seen that Admittance Control results in oscillations during stiff contact (due to time delay in the force measurement and the lag of the position controller) but has good position accuracy in free space. In contrast, Impedance Control has negligible oscillations in stiff contact but has larger steady state error in free space.

Figure 25 shows the result of our hybrid controller with $\delta = 20$ ms for different values of duty cycle. It can be seen that there is no oscillation in stiff contact. In free space, the steady state error is less than that of Impedance Control and decreases for higher duty cycles. The results clearly demonstrate the possibility of using Hybrid Impedance and Admittance Control for improving upon the steady state performance

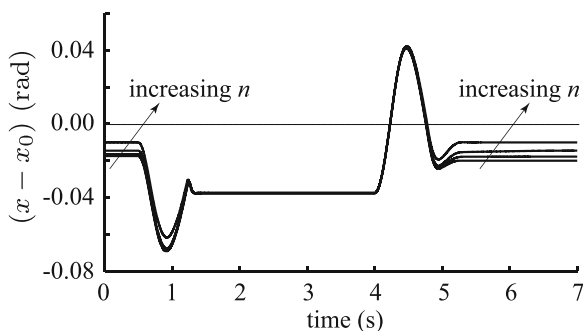


Fig. 25 Experimental result with Hybrid Impedance and Admittance Control

of Impedance Control and stability characteristics of Admittance Control.

6 Conclusion

We presented a new solution to the impedance control problem in which we continuously switch between controllers with impedance and admittance causality. By taking the duty cycle as a design parameter, we arrive at a family of controllers whose stability and performance characteristics interpolate those of classical Impedance Control and Admittance Control. Using both numerical simulations and experiments we have shown that this approach allows us to effectively combine the robustness property of Impedance Control in stiff contact with the accuracy of Admittance Control in soft contact. The analysis, simulations, and experiments in this paper were all based on a single degree-of-freedom system, and naturally, extension of the approach to nonlinear, coupled, multi-degree-of-freedom systems is an important goal of our future research. Through numerical simulations it was shown that the hybrid controller has the potential to provide the best performance when the duty cycle is adapted to the environment stiffness. Thus, another important research problem is to develop an adaptive algorithm for on-line estimation of the environment stiffness and adaptation of the duty cycle. Finally, in this work we assumed that our physical system remains in contact with the environment at all times and there are no contact transitions. In reality, contact transitions may occur and such transitions will be accompanied by impulsive forces. The mechanics of contact transitions has been studied extensively in the literature (see [4], for example) and future work on impedance control should estimate the velocity of impact and include the effect of the impulsive forces resulting from contact transitions.

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